

Inflation Inequality Across the Euro Area

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Motivation, Research Question, and Literature

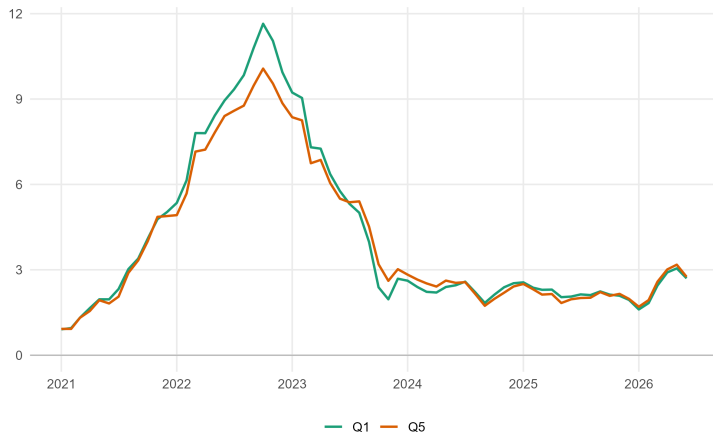
- ▶ Central banks target aggregate inflation, but households consume different baskets.
- ▶ The 2021–2023 episode combined high aggregate inflation with large relative-price shocks in energy and food.
- ▶ How different was inflation across income, age, and residence groups in the euro area?
- ▶ Which products and production-network channels generated these differences?
- ▶ Did household price policies reduce average inflation and inflation inequality?
- ▶ The paper connects research on household-specific inflation, distributional inflation channels, energy exposure, and unconventional fiscal policy.
- ▶ Related work includes Doepke and Schneider; Kaplan and Schulhofer-Wohl; Jaravel; Pallotti et al.; Claeys et al.; Levell et al.; and Gautier et al.

Data and Methodology

- ▶ Monthly Eurostat HICP price indices and item weights, harmonized across euro-area countries.
- ▶ Aggregate HBS tabulations provide expenditure shares by income, age, residence, and COICOP category.
- ▶ Individual HBS microdata provide household expenditures, disposable income, demographics, and survey weights.
- ▶ HBS expenditures are harmonized with HICP items through a COICOP bridge.
- ▶ Group-specific expenditure weights are estimated using RAS balancing.
- ▶ Country-level chained Laspeyres indices are aggregated using annual HICP country weights.

$$P_{g,t} = P_{g,t-1} \sum_k w_{gk,t-1} \frac{p_{k,t}}{p_{k,t-1}}, \quad \pi_{g,t} = 100 \left(\frac{P_{g,t}}{P_{g,t-12}} - 1 \right).$$

Short-Run Inflation Inequality by Income Quintile



Source: Eurostat HICP and HBS; authors' calculations.

Inflation Inequality by Age



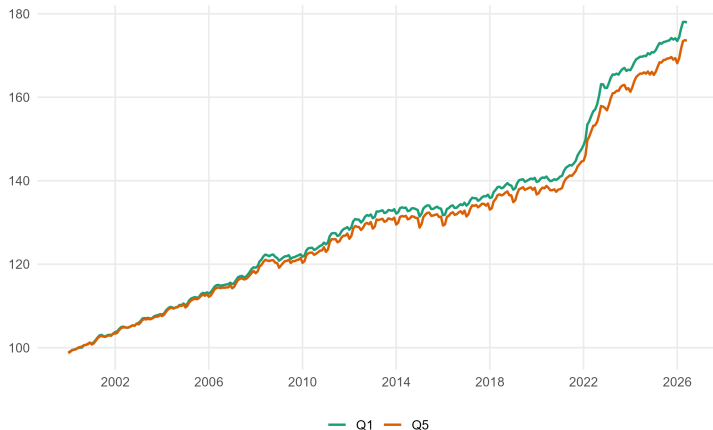
Source: Eurostat HICP and HBS; authors' calculations.

Inflation Inequality by Residence Area



Source: Eurostat HICP and HBS; authors' calculations.

Long-Run Inflation Inequality by Income Quintile



Source: Eurostat HICP and HBS; authors' calculations.

Cumulative Inflation Gap Across Countries, June 2021–June 2023

Country	Total inflation	Q1 inflation	Q5 inflation	Gap
Austria	17.2	17.5	17.1	0.4
Cyprus	12.0	12.7	12.0	0.7
Estonia	33.0	38.7	31.2	7.5
Finland	12.5	11.5	12.7	-1.2
France	12.2	12.3	12.1	0.2
Germany	15.6	16.5	15.2	1.3
Ireland	14.9	17.2	13.3	3.8
Italy	15.8	17.2	15.1	2.1
Latvia	28.9	36.0	26.5	9.5
Lithuania	30.4	33.6	29.0	4.6
Netherlands	16.9	17.7	16.5	1.2
Portugal	14.2	13.1	15.4	-2.3
Slovakia	25.4	26.7	24.0	2.7
Spain	11.8	11.8	11.7	0.1
Euro area	14.7	15.2	14.4	0.9

Source: Eurostat HICP and HBS; authors' calculations. Percentage points.

Product Contributions to the Income Inflation Gap

Product group	Headline	Q1	Q5	Gap
Food and non-alcoholic beverages	4.3	5.2	4.5	0.7
Alcohol, tobacco and narcotics	0.5	0.7	0.4	0.2
Clothing and footwear	0.3	0.3	0.4	-0.1
Housing, rentals and repairs	1.2	0.4	0.6	-0.2
Water, electricity, gas and fuels	1.5	6.9	5.3	1.5
Transport	1.9	1.6	2.8	-1.2
Other	5.0	0.2	0.3	-0.1
Total	14.7	15.2	14.4	0.9

- Energy and food increase the Q1–Q5 gap; transport exposure offsets much of it.

Source: Eurostat HICP and HBS; authors' calculations. Percentage points.

Direct and Indirect Exposure to an Oil and Gas Shock

Country	Total	Direct	Indirect	Gap	Gap direct	Gap indirect
Austria	11.8	3.9	7.9	-1.4	-0.7	-0.7
Belgium	9.8	6.2	3.7	-3.6	-3.2	-0.4
Germany	7.4	4.4	3.0	-1.8	-1.6	-0.2
Estonia	12.2	6.6	5.6	-4.8	-3.8	-1.0
Spain	7.4	4.5	3.0	-1.9	-1.6	-0.3
France	9.3	4.5	4.7	-2.8	-1.9	-0.9
Italy	9.2	4.9	4.3	-1.9	-1.3	-0.6
Latvia	14.1	7.2	6.9	-7.9	-4.9	-3.0
Netherlands	6.4	4.3	2.0	-1.4	-1.3	0.0
Portugal	8.8	3.7	5.2	-4.3	-2.1	-2.2
Slovakia	12.5	6.3	6.2	-0.5	-0.4	-0.1
Euro area	8.4	4.6	3.8	-2.2	-1.7	-0.5

- Uniform 40% shock; price propagation is computed as $\Delta p = (I - A')^{-1}s$.

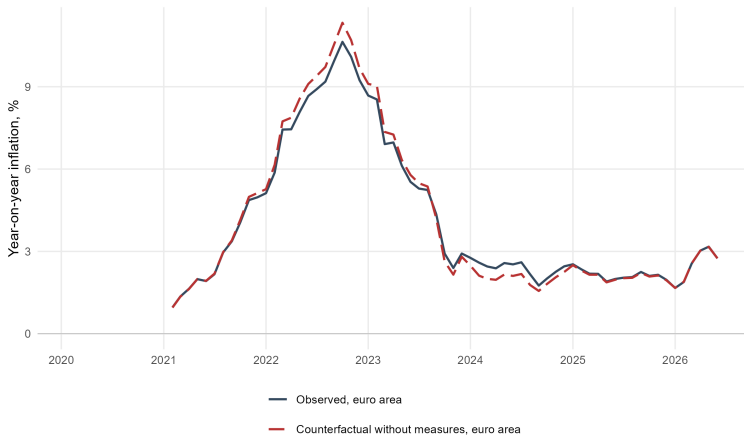
Source: Eurostat HICP, HBS and FIGARO; authors' calculations.

Unconventional Household Price Policies

- ▶ Electricity and gas price caps or tariff shields limited the retail prices paid by households.
- ▶ Temporary VAT and excise-tax reductions lowered consumer energy prices.
- ▶ Cuts in regulated network charges reduced electricity and gas bills.
- ▶ Fuel rebates and regulated discounts reduced prices at the pump.
- ▶ The counterfactual removes these price-based interventions while retaining observed prices for unaffected products.
- ▶ Direct income transfers and measures targeted only at firms are excluded from the price counterfactual.
- ▶ Policy effects are measured as observed inflation minus estimated no-policy inflation.

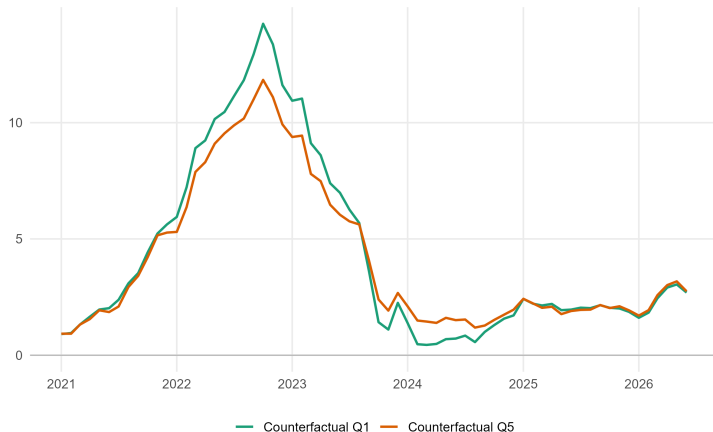
Source: National policy sources, Sgaravatti et al.; authors' calculations.

Household Price Policies and Average Inflation



Source: Eurostat and national statistical institutes; authors' calculations.

Household Price Policies and Inflation Inequality



Source: Eurostat and national statistical institutes; authors' calculations.

Fiscal Cost and No-Policy Inflation Across Countries

Country	Cost	No-policy	Avg. effect	Q1	Q5	Gap	Gap effect
Austria	3.8	17.7	-0.5	19.3	18.2	1.1	-0.7
Estonia	0.2	33.1	-0.1	39.0	31.4	7.6	-0.1
France	80.5	14.0	-1.8	19.4	16.6	2.8	-2.6
Germany	60.2	16.0	-0.4	18.4	16.2	2.2	-0.9
Ireland	1.4	15.0	-0.1	17.6	13.6	4.1	-0.2
Italy	36.2	16.1	-0.3	18.2	15.8	2.4	-0.3
Latvia	0.7	29.2	-0.2	37.2	27.0	10.2	-0.6
Netherlands	34.8	17.6	-0.7	20.3	18.0	2.3	-1.1
Portugal	5.6	14.8	-0.6	15.5	17.0	-1.5	-0.8
Slovakia	2.3	26.9	-1.6	33.5	27.8	5.7	-3.0
Spain	23.1	12.5	-0.7	14.6	13.5	1.1	-1.0
Euro area	266.9	15.4	-0.7	18.2	16.2	2.0	-1.2

- Negative effects indicate that price policies reduced inflation or the Q1–Q5 gap.

Source: Eurostat and national statistical institutes; authors' calculations.

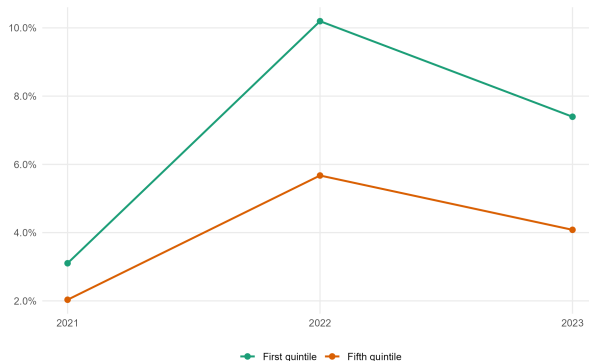
Product-Level Effects of Household Price Policies

Product group	No-policy	Q1	Q5	Gap	Policy effect
Food and non-alcoholic beverages	4.5	5.4	4.6	0.8	-0.1
Alcohol, tobacco and narcotics	0.5	0.7	0.4	0.2	0.0
Clothing and footwear	0.3	0.3	0.4	-0.2	0.0
Housing, rentals and repairs	0.5	0.4	0.6	-0.2	0.0
Water, electricity, gas and fuels	6.9	9.4	6.5	2.9	-1.4
Transport	2.4	1.8	3.2	-1.4	0.2
Other	0.3	0.2	0.4	-0.2	0.0
Total	15.4	18.2	16.2	2.0	-1.2

- Energy accounts for almost all of the reduction in inflation inequality.

Source: Eurostat and national statistical institutes; authors' calculations.

Household-Level Inflation Burden



- ▶ Low-income households face a larger inflation burden relative to net income.
- ▶ Price policies reduced both average inflation and the income inflation gap.

Source: Eurostat HICP and HBS microdata; authors' calculations.

Conclusion

- ▶ Inflation differences between income groups are moderate for the euro area as a whole, but vary substantially across countries.
- ▶ Age and residence-area gaps were larger than the aggregate Q1–Q5 gap during the 2021–2023 inflation episode.
- ▶ Food and household energy increased inflation exposure for low-income households, while transport partly offset the income gap.
- ▶ Production networks transmitted energy shocks indirectly across sectors and countries.
- ▶ Household price policies lowered cumulative average inflation and reduced the Q1–Q5 inflation gap.
- ▶ Inflation remained more burdensome for low-income households because consumption represents a larger share of their net income.